

ANSHUMAN NAYAK

Senior Full-Stack Engineer · AI/ML Engineer

anshuman_n@outlook.com | +91 9337928411 | Bengaluru, India | anshumann.com

PROFESSIONAL SUMMARY

Full-Stack Engineer and AI/ML specialist with 6 years of experience building high-scale production systems across healthcare, agri-tech, and e-commerce. Expertise spans LLM-powered products (GPT, Claude, Gemini), multi-agent orchestration, real-time ETL pipelines, payment infrastructure, and domain-specific AI evaluation. IIT Kharagpur postgraduate with 7 peer-reviewed publications (129+ citations), 1 granted patent in smartphone-based soil sensing, and proven leadership mentoring cross-functional teams.

EXPERIENCE

Zeno Health · Software Engineer II | Remote | Feb 2024 – Present

- Engineered high-performance payment gateway processing 15,000+ daily transactions with 99.9% uptime and sub-100ms latency, supporting millions in monthly transaction volume.
- Architected scalable ETL pipelines managing monthly migration of 10B+ records with zero data loss, optimizing data integrity, consistency, and processing throughput across multiple databases.
- Deployed enterprise-scale NLP chatbots powered by GPT-4 handling 1M+ customer inquiries annually, reducing manual support volume by 60% while maintaining quality standards.
- Orchestrated autonomous multi-agent systems using GPT-4, Gemini-2, and Claude Sonnet to automate complex product development workflows, improving team efficiency by 70%.
- Led and mentored cross-functional team of 10 developers across full SDLC, establishing code quality standards, implementing self-healing AI agents for automated code review and proactive bug detection.

Carelon Global Solutions LLP · Software Engineer | Gurugram, India | Jul 2022 – Feb 2024

- Led creation of low-code Data API Gateway platform, reducing backend development time from 14 days to 2 hours via reusable templates and intelligent code generation.
- Engineered fault-tolerant, scalable core infrastructure supporting 2,000+ data scientists and 5+ production applications simultaneously with 99.9% uptime.
- Managed end-to-end data model lifecycle including schema design, pipeline optimization, and implemented comprehensive alerting systems for real-time monitoring and incident response.

NexVerse Pvt. Ltd. · Full-Stack Developer | Bhubaneswar, India | Jan 2019 – Mar 2022

- Developed Cordova-based hybrid social media application serving 8,000+ users across 15+ businesses with real-time features and seamless cross-platform experience.
- Built Node.js and MongoDB backend infrastructure with push notifications, in-app messaging, A/B testing capabilities, and integrated analytics for continuous product improvement.
- Implemented AWS EC2 autoscaling and optimized video processing pipeline, reducing latency by 40% and improving overall user experience quality.

SKILLS

Languages: Python, JavaScript, TypeScript, Go, C | **AI/LLM:** GPT, Claude, Gemini, LangGraph, Multi-Agent Systems, RAG, LangChain

Frameworks: Node.js, Express, React, Next.js, Django, Flask, GraphQL, REST API, TensorFlow, OpenCV

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Firebase, Snowflake, AWS (EC2, EKS, RDS, S3, Amplify), Docker, Nginx

EDUCATION

IIT Kharagpur | M.Tech, Agricultural Systems & Management | CGPA: 9.51/10 | All India Rank 67 — GATE 2020

Orissa University of Agriculture and Technology | B.Tech, Agricultural Engineering | CGPA: 8.50/10

PROJECTS

Fusion CMS — TypeScript, Next.js, GraphQL, REST, SQL/NoSQL

Auto-generates high-quality GraphQL and REST backends from SQL/NoSQL sources with zero explicit coding, cutting backend development time significantly. Deploys as monolith or microservice with middleware customization and Next.js frontend for maximum scalability.

SpectraML — Django, Python, Plotly, D3.js, Jupyter

Open-source AutoML platform for spectroscopic data analysis — automates preprocessing, feature engineering, model selection, PCA/t-SNE visualizations, and evaluation. Full-stack application with real-time experimentation and reproducible Jupyter pipelines.

Mobile Plant Disease Detector — JavaScript, Python, TensorFlow.js, WebAssembly, Ionic

Offline-capable mobile application enabling Indian farmers to identify plant diseases via smartphone camera using deep learning. Combines TensorFlow.js and OpenCV.js with mobile-first architecture. Research published in Elsevier.

Haiti Digital Soil Mapping — Next.js, Python FastAPI, MySQL

Country-wide location-specific digital soil mapping platform providing available nutrient analysis and precision fertilizer recommendations. Deployed platform used by policymakers, fertilizer companies, and smallholder farmers for data-driven farm management.

CONSULTANCY

Micro1 AI · Human Data Manager | Remote | Aug 2025

- Leading and scaling team of 53 Subject Matter Experts across Data Science, Healthcare, Physics, and Executive Leadership domains for high-quality data generation.
- Curating and validating high-quality full-stack code repositories (Node.js, React, Python, Go, Ruby) to optimize LLM training datasets and improve model reasoning capabilities.
- Engineering organic data ingestion pipelines prioritizing human-generated content over synthetic data, contributing to improved model alignment and AGI development.
- Establishing rigorous multi-stage data quality review cycles and human-in-the-loop workflows to enhance model alignment, code fidelity, and evaluation robustness.

Scale AI · Human Frontier Collective Fellow | Engineering Expert | Jan 2025 – Aug 2025

- Evaluated and assessed AI model solutions across agricultural and environmental engineering domains, rigorously identifying errors, limitations, and failure modes.
- Curated domain-specific test questions and evaluation benchmarks to systematically assess AI model capabilities and reliability in agricultural technology applications.
- Provided expert domain feedback and correct technical solutions for identified model mistakes, directly contributing to improved AI training data quality and model evaluation datasets.
- Leveraged deep agricultural systems knowledge to guide AI development toward more robust, reliable solutions in soil science, precision farming, and environmental monitoring.

PUBLICATIONS & HONORS

Publications: 7 peer-reviewed papers with 129+ citations across leading journals including Smart Agricultural Technology, Microchemical Journal, Sensors, Soil Advances, Scientific Reports, and Geoderma Regional. Research focuses on smartphone-based soil sensing, digital soil mapping, proximal sensing technologies, and generative AI applications.

Awards & Recognition: Patent granted for smartphone-based detection of nitrate and phosphate in water and soil samples | Recognized Scholar in Digital Agricultural Technology (Geological Society of America, Kyoto, Japan) | All India Rank 67 in GATE 2020 (Agricultural Engineering)

Certifications: GraphQL Developer Associate (Apollo) | Meta Back-End Developer Program | Data Analytics with Python - Silver Elite Award (NPTEL) | JavaScript Algorithms and Data Structures (FreeCodeCamp) | Responsible and Safe AI (NPTEL)